

Pension Justice Lab — One-Page Overview

Purpose

Pension Justice Lab quantifies how gender, racial, and caregiving gaps reshape retirement outcomes and system solvency. It turns “equity” into solvency math—estimating replacement rates, poverty risk, and intergenerational effects under different policy levers (caregiver credits, top-ups, portable benefits). Models blend OECD/WEF/IFoA parameters with your inputs to produce clear comparisons and a short analytic note explaining what drives the gap.

Core outputs - Replacement rate (% of final earnings) - Lifetime poverty probability - Solvency impact (years of funding horizon gained/lost) - Intergenerational contribution shift - Policy cost vs. solvency gain

Intended Users

- **Pension trustees & CIOs** — test inclusion reforms for funding impact.
- **Policy analysts & regulators** — pre-legislate caregiver credits, top-ups, and auto-portability.
- **Institutional investors** — treat social gaps as quantifiable risk variables.
- **Researchers & journalists** — visualize the cost of inequity with defensible numbers.

How to Use It (3 steps)

- 1) **Provide a profile**
 - Country/system, age, current earnings, contribution rate
 - Career interruptions (years), gig/portable top-ups, retirement age
 - Optional: sector, income decile, expected return/fee assumptions
- 2) **Choose scenarios to compare**
 - Baseline vs **Caregiver Credits** vs **Portable Wallet** vs **Both**
 - **Climate Longevity Stress** (1.5–3 °C pathways)
 - **Jurisdictional Comparison** (e.g., Canada, Netherlands, Australia, Singapore)
 - **Board Readiness Score** (five-question governance audit)
- 3) **Read the dashboard**
 - One bar-chart comparison + a short analytic note (e.g., “80% of the gap is missed contribution years; 20% differential returns”).

Copy-paste starter prompts - “Model a 38-year-old woman in Canada earning \$70k with a 6-year caregiving break; compare baseline vs. Germany-style caregiver credit.” - “Simulate a gig worker in Canada earning \$45k with a 5% platform top-up; show solvency and poverty outcomes.” - “Run a 2 °C climate stress for a female care-sector worker in the Netherlands.” - “Compare the same worker across Canada, Australia, and Singapore.”

What’s Inside the Math (conceptual)

$$[\text{pension_balance}[t+1] = \text{pension_balance}[t] + (\text{income}[t] \times \text{contrib_rate}) + (\text{pension_balance}[t] \times \text{return_rate})]$$

Modifiers: career breaks reduce contributions; caregiver credits add phantom earnings; portable wallets add platform co-contributions; mortality varies by gender, income decile, and region.

Limitations & Ethics

- **Estimates, not advice.** Results are indicative ($\pm 10\%$ typical) and depend on assumptions.
- **Policy simplification.** Country rules vary; niche provisions or pending reforms may be omitted.
- **Data coverage.** Uses OECD/WEF/IFoA baselines and stylized mortality tables; local data may shift results.
- **No demographic inference.** The model never infers protected traits; only uses what you explicitly provide.
- **Privacy.** Inputs are used to generate scenarios; do not share sensitive personal data.

Creator

Tanya Matanda — governance analyst and writer focused on fiduciary accountability and actuarial realism. Author of the *Pension Systems at the Edge* series and work on fiduciary capital, climate solvency, and AI-ready pension governance. Builds policy models, boardroom visuals, and comparative benchmarks that treat fairness as solvency hygiene.

Link to this GPT

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